

CS319: Scientific Computing

Vector and Matrix classes

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Slides and examples:

<https://www.niallmadden.ie/2526-CS319>



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1. Overview

This week is all about extending the `Vector` class from Week 9, which were stored in files called `VectorBasic.h` and `VectorBasic.cpp`.

Since the code for the header and definition are different from last week's version, the new files have different names: `Vector10.h`, `Vector10.cpp`.

Previously, we that (e.g.) the `+` operator was “overloaded” to allow us to “add” (i.e., concatenate) two strings. We want to see how overload operators for classes that we write so that, for example, we can use the `+` operator to add two vectors.

That is, we want to study **Operator Overloading**. But to get this to work, we need to study **copy constructors**.

This is a technical area of C++ programming, but is unavoidable.

As we already know, **constructor** is a method associated with a class that is called automatically whenever an object of that class is declared.

But there are time when objects are *implicitly* declared, such as when passed (by value) to a function.

Since this will happen often, we need to write special constructors to handle it.

Last week we defined a class for vectors:

- ▶ It stores a vector of N doubles in a dynamically assigned array called *entries*;
- ▶ The constructor takes care of the memory allocation.

```
2 // From VectorBasic.h (Week 9)
3 class Vector {
4     private:
5         double *entries; ← pointer
6         unsigned int N;
7     public:
8         Vector (unsigned int Size=2);
9         ~Vector(void);
10
11         unsigned int size(void) {return N;};
12         double geti (unsigned int i);
13         void seti (unsigned int i, double x);
14         // print(), zero() and norm() not shown
15 };
16
17 // Code for the constructor from Vector.cpp
18 Vector::Vector (unsigned int Size) {
19     N = Size;
20     entries = new double[Size];
21 }
```

memory allocation

We then wrote some functions that manipulate vectors, such as `AddVec` in `Week09/00TestVectorBasic.cpp`

```
2 void VecAdd (Vector &c, Vector &a, Vector &b,  
              double alpha=1.0, double beta=1.0);
```

Note that the `Vector` arguments are passed by reference...

&

sets

$$c = \alpha a + \beta b$$

α, β are scalars (ie floats)

a, b, c are Vectors

What would happen if we tried the following, seemingly reasonable piece of code?

```
Vector x(4);  
x.zero(); // sets entries of x all to 0  
Vector y=x; // should define a new vector, with a copy of x
```

This will cause problems for the following reasons:

This will set $y.N = x.N$ (OK!)

and $y.entries = x.entries$ (not OK!)

These now share a memory address,
which can be problematic....

Problems

- ① Changing (e.g.) `y.entries[0]`
would change `x.entries[0]`
Probably not intended.
- ② Worse, when (e.g.) `y` goes
out of scope, the destructor is
called, and the memory allocated
to `y.entries` is deleted/deallocated.
So `x.entries` is deallocated too.

Worse still, x will eventually go out of scope.

when its destructor is called it will call a "delete" for $x.entries$, which is already deleted. So the program will crash ("double delete").

To solve this problem, we should define our own **copy constructor**, which makes a copy of an existing object. Therefore, it takes a single parameter: the address of the object to copy. For example:

See Vector10.cpp for more details

```
20 // copy constructor
   Vector::Vector (const Vector &old_Vector)
22 {
   N = old_Vector.N;
24   entries = new double[N];
   for (unsigned int i=0; i<N; i++)
26     entries[i] = old_Vector.entries[i];
   }
```

makes a new copy of the values of old_Vector.entries

The **copy constructor** can be called two ways:

(a) **explicitly**, .e.g,

```
Vector V(2);  
V.set1(0)=1.0; V.set1(1)=2.0;  
Vector W(V); // W is a copy V
```

(b) More commonly: **implicitly**, when ever an object is passed by value to a function. If we have not defined our own copy constructor, the default one is used, which usually causes trouble.

In this section, we'll study **“Operator overloading”**.

Our main goal is to overload the addition (+) and subtraction (-) operators for vectors.

Last week, we wrote a function to add two **Vectors**: **AddVec**.

It is called as **AddVec(c,a,b)**, and adds the contents of vectors *a* and *b*, and stores the result in *c*.

It would be much more natural to redefine the standard **addition** and **assignment** operators so that we could just write **c=a+b**. This is called **operator overloading**.

To overload an operator we create an **operator function** – usually as a member of the class. (It is also possible to declare an operator function to be a **friend** of a class – it is not a member but does have access to private members of the class. More about **friends** later).

The general form of the operator function is:

```
return-type class-name::operator#(arguments)  
{  
    :    // operations to be performed.  
};
```

return-type of a operator is usually the class for which it is defined, but it can be any type.

Note that we have a new key-word: **operator.**

The operator being overloaded is substituted for the # symbol

Almost all C++ operators can be overloaded:

+	-	*	/	%	^	&		~	!
=	<	>	+=	-=	<u>*=</u>	<u>/=</u>	<u>%=</u>	^=	&=
=	<<	>>	>>=	<<=	<u>==</u>	<u>!=</u>	<=	>=	&&
	++	--	->*	,	->	[]	()	new	delete

but not

. :: .* ?

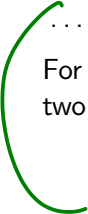
- ▶ Operator precedence cannot be changed: `*` is still evaluated before `+` : $a = b + c * d$ is always $a = b + (c * d)$
- ▶ The number of arguments that the operator takes cannot be changed, e.g., the `++` operator will still take a single argument, and the `/` operator will still take two.
- ▶ The original meaning of an operator is not changed; its functionality is extended. It follows from this that operator overloading is always relative to a user-defined type (in our examples, a `class`), and not a built-in type such as `int` or `char`.
- ▶ Operator overloading is always relative to a user-defined type (in our examples, a `class`).
- ▶ The assignment operator `=` is automatically overloaded, but in a way that usually fails except for very simple classes.

We are free to have the overloaded operator perform any operation we wish, but it is good practice to relate it to a task based on the traditional meaning of the operator. E.g., if we wanted to use an operator to add two matrices, it makes more sense to use `+` as the operator rather than, say, `*`.

We will concentrate mainly on binary operators, but later we will also look at overloading the unary “minus” operator.

.....

For our first example, we'll see how to overload `operator+` to add two objects from our `Vector` class.



$a = b - c$ (binary)
 $a = -b$ (unary)

First we'll add the declaration of the operator to the class definition in the header file, [Vector10.h](#):

See Vector10.h for more details

```
14  class Vector {  
15      friend std::ostream &operator<<(std::ostream &, Vector &v);  
16  private:  
17      double *entries;  
18      unsigned int N;  
19  public:  
20      Vector (unsigned int Size=2);  
21      Vector (const Vector &v); // Arg must be passed by value. Why?  
22      ~Vector(void);  
23      Vector operator+(Vector b); // add two vectors
```

ignore for now!!

*and return their sum
as a vector.*

Then to `Vector10.cpp`, we add the code

See `Vector10.cpp` for more details

```
96 // Overload the + operator.  
96 Vector Vector::operator+(Vector b)  
96 {  
98     Vector c(N); // Make c the size of a  
98     if (N != b.N)  
100         std::cerr << "vector::+ cannot add 2 vecs of different size!"  
100         << std::endl;  
102     else  
102         for (unsigned int i=0; i<N; i++)  
104             c.entries[i] = entries[i] + b.entries[i];  
106     return(c);  
106 }
```

If we call this as $c = a + b$, then $N = a.N$, etc.

96: It looks like $+$ has only one argument!
When we call $a+b$, this is essentially $a.operator+(b)$.
So left argument is implicit!

First thing to notice is that, although `+` is a binary operator, it seems to take only one argument. This is because, when we call the operator, `c = a + b` then `a` is passed **implicitly** to the function and `b` is passed **explicitly**.

Therefore, for example, `a.N` is known to the function simply as `N`.

The temporary object `c` is used inside the object to store the result. It is this object that is returned. Neither `a` or `b` are modified.

We now want to see another way of accessing the implicitly passed argument. First, though, we need to learn a little more about pointers, and introduce a new piece of C++ notation.

Recall that if, for example, `x` is a `double` and `y` is a pointer to `double`, we can set `y=&x`. So now `y` stores the memory address of `x`. We then access the contents of that address using `*y`.

Now suppose that we have an object of type `Vector` called `v`, and a *pointer to `vector`*, `w`. That is, we have defined

```
2  Vector v;  
    Vector *w;
```

Then we can set `w=&v`. Now accessing the member `N` using `v.N`, will be the same as accessing it as `(*w).N`.

It is important to realise that `(*w).N` is **not** the same as `*w.N`.

C++ provides a new operator for this situation: `w->N`, which is equivalent to `(*w).N`.

$*w.N$ is equivalent to $*(w.N)$

But w is a pointer, so has
no member called N (bug!)

Finish here Wednesday

When writing code for functions, and especially overloaded operators, it can be useful to **explicitly** access the implicitly passed object.

That is done using the `this` pointer, which is a pointer to the object itself.

.....

As we've just noted, since `this` is a pointer, its members are accessed using either `(*this).N` or `this->N`.

We often use the `this` pointer when a function must return the address of the argument that was passed to it. This is the case of the assignment operator.

See Vector10.cpp for more details

```
100 // Overload the = operator.  
101 Vector &Vector::operator=(const Vector &b)  
102 {  
    if (this == &b)  
        return(*this); // Taking care for self-assignment  
  
    delete [] entries; // In case memory was already allocated  
  
    N = b.N;  
    entries = new double[b.N];  
    for (unsigned int i=0; i<N; i++)  
        entries[i] = b.entries[i];  
  
    return(*this);  
}
```

Used in
 $a = b$

↑
implicit.

note
 $b = b$ is ok.

5. Unary Operators

So far we have discussed just the **binary** operator, `+`. By “**binary**”, we mean it takes **two** arguments.

But many C++ operators are **unary**: they take only one argument; examples include `++` and `--`.

For our `Vector` class, we want to overload the `-` (minus) operator. Note that this can be used in two ways:

▶ $c = (-a)$ (unary).

▶ $c = (a - b)$ (binary)

*argument / input is
to the right of the
operator.*

In the first case here, “minus” is an example of a **prefix** operator. (See “**Extras**” for an example of overloading **postfix** operators, like `a++`, which are a little more complicated).

After that we will then define the binary minus operator, by using addition and unary minus.

For the unary “minus” operator, when we write “-a” the object `a` is passed *implicitly*.

This is a little different from previous cases, where the object passed implicitly is to the left of the operator.

See `Vector10.cpp` for more details

```
108 // Overload the unary minus (-) operator. As in b=-a;  
Vector Vector::operator-(void) because unary.  
110 {  
    Vector b(N); // Make b the size of a  
112    for (unsigned int i=0; i<N; i++)  
        b.entries[i] = -entries[i];  
114    return(b);  
}
```

And now that we have defined this operator, we can define the **binary** minus operator. Now this time when we write “a-b”, it is the *left* argument that is implicit.

See Vector10.cpp for more details

```
118 // Overload the binary minus (-) operator. As in c=a-b
119 // This implementation reuses the unary minus (-) operator
120 Vector Vector::operator-(Vector b)
121 {
122     Vector c(N); // Make b the size of a
123     if (N != b.N)
124         std::cerr << "Vector:: operator- : dimension mismatch!"
125                     << std::endl;
126     else
127         c = *this + (-b);
128     return(c);
129 }
```

do this as $c = a + (-b)$

reusing unary - for prev. slide.

re-use binary + from earlier

6. friend functions

In all the examples that we have seen so far, the only functions that may access private data belonging to an object has been a member function/method of that object.

If we need a function that does not belong to the class to be able to access **private** elements, it can be designated a **friend** of the class.

For non-operator functions, there is nothing that complicated about **friends**. However, care must be taken when overloading operators as **friends**.

In particular:

- ▶ All arguments are passed explicitly to **friend** functions/operators.
- ▶ Certain operators, particularly the **insertion/put-to** **<<** and **extraction/get-from** **>>** operators can only be overloaded as friends.

In last week's version of the `Vector` class, we could output its elements using the `print()` method. E.g.:

```
Vector v;  
2 v.zero()  
std::cout << "v  has values  ";  
4 v.print();
```

But it would be much more convenient just to do

```
std::cout << "v  has values  " << v;
```

But the `insertion` operator was not defined for our class.

We can fix that, by overloading it. However, the `<<` operator belongs to `std::cout`, not to `Vector`. So it cannot access its `entries` member.

Here is how we resolve this...

We add the following line to the definition of the `Vector` class.

```
friend std::ostream &operator<<(std::ostream &, Vector &v);
```

And then we define:

↑ in `VectorIO.h`. *Simplicit (usually cout)*

```
1 std::ostream &operator<<(std::ostream &output, Vector &v)
2 {
3     output << "[";
4     for (unsigned int i=0; i<v.size()-1; i++)
5         output << v.entries[i] << ",";
6     output << v.entries[v.size()-1] << "]";
7
8     return(output);
9 }
```

Now we can display a vector using `std::cout` directly.

output is acting like cout, or like a file pointer.

7. Testing the `Vector` class

I've provided a program called `01TestVectorOperators.cpp` which tests various operators that we have defined.

To run that, you'll need a **project** that contains all three files. If you must use an online compiler (though I advise against it), I suggest trying online-cpp.com.

8. Preprocessor Directives

Our next step is to define a `Matrix` class, and overload some of the associated operators. One of those is the multiplication (“times”) operator `*` for matrix-vector multiplication.

With those done, we can think about overloading the multiplication operator for *Matrix-Vector* multiplication.

This introduces a few small new complications:

- ▶ the return type is different from the class type;
- ▶ if we use multiple source files, how do we know where exactly to place the `#include` directives?

So, before we can proceed, we need to take a short detour to consider **preprocessor** directives.

The preprocessor in C++ is a hang-over over from early versions of C. Originally, that language did not have a construct for defining constants and including header files. To get around this, an early version of C introduced the **preprocessor**. This is a program that

- ▶ reads and modifies your source code by checking for any lines that being with a hash symbol (**#**);
- ▶ carries out any operations required by these lines;
- ▶ forms a new source code that is then compiled.

We usually don't get to see this new file, though you can view it by compiling with certain options (with **g++**, this is **-E**).

The preprocessor is *separate* from the compiler, and has its own syntax.

The simplest preprocessor directive is `#define`. This is used for defining global constants, and doing a simple search-and-replace. For example,

```
#define SIZE 10
```

will find every instance of the word (well, token, really) *SIZE* and replaces it with *10*.

In general, this use of the `#define` directive to define identifiers to be used like “global variables” is not very good practice. However, it can be very useful as a way of checking if a piece of code has already been compiled.

The most familiar preprocessor is `#include`, e.g.,

```
1 #include <iostream>
   #include "Vector10.h"
```

This tells the preprocessor to take the named file(s) and insert them into the current file.

If the name is contained in angle brackets, as in `iostream`, this means the preprocessor will look in “the usual place” – where the compiler is installed on your system.

If the named file is in quotes, it looks in the current directory/folder, or in the specified location.

Finally, we have **conditional compilation**.

Suppose we want to write a member function for the *Matrix* class that involves the *Vector* class.

So we need to include *Vector10.h* in *Matrix10.h*. But then if our main source file includes both *Matrix10.h* and *Vector10.h* we could end up defining it twice.

To get around this we use *conditional compilation*.

In the files we can have such lines as the following in *Vector10.h*

```
2 #ifndef _VECTOR_H_INCLUDED
  // stuff goes here
4 #endif
```

ifndef means
"if not defined".

9. A matrix class

We now write a `class` implementation for a `matrix`, along with the associated functions.

We'll first consider how the matrix is data is stored. The most natural approach might seem to be to construct a two dimensional array. This can be done as follows:

```
double **entries = new double *[N];  
for (int i=0; i<N; i++)  
    entries[i] = new double N;
```

A simpler, faster approach is to store the N^2 entries of the matrix in a single, one-dimensional, array of length N^2 , and then take care how the access is done:

$\hookrightarrow A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ stored as $A = [1, 2, 3, 4, 5, 6, 7, 8, 9]$
 $A[i, j]$ $\rightarrow A[i * N + j]$

Handwritten notes: Red circled 6 in the matrix, red circled 6 in the array, and red $2N+0$ above the array index 6.

9. A matrix class

Matrix10.h

```
12 class Matrix {  
    private:  
14     double *entries;  
        unsigned int N;  
16 public:  
    Matrix (unsigned int Size=2);  
18     Matrix (const Matrix &m); // Copy constructor  
        ~Matrix(void);  
  
    Matrix &operator=(const Matrix &B); // assignment operator  
  
    unsigned int size(void) {return (N);};  
24     double getij (unsigned int i, unsigned int j);  
        void setij (unsigned int i, unsigned int j, double x);  
  
    Vector operator*(Vector u); // Define later!  
28     void print(void);  
};
```

9. A matrix class

First we'll look at the code for the constructor, to verify that the data is stored just as a 1D array:

from [Matrix10.cpp](#)

```
12 // Basic constructor. See below for copy constructor.  
12 Matrix::Matrix (unsigned int Size)  
13 {  
14     N = Size;  
15     entries = new double [N*N];  
16 }
```

9. A matrix class

Next we'll look at the `setij()` member, to see how indexing works.

from `Matrix10.cpp`

```
24 void Matrix::setij (unsigned int i, unsigned int j, double x)
   {
26     if (i<N && j<N)
        entries[i*N+j]=x;
28     else
        std::cerr << "Matrix::setij(): Index out of bounds.\n";
   }
```

Other components of the `Matrix` class are similar to the corresponding functions for the `Vector` class, such as the assignment operator, and the copy constructor.

So, we'll just focus on overloading the `operator*` for multiplication of a vector by a matrix: `c=A*b`, where A is an $N \times N$ matrix, and c and b are vectors with N entries.

Since the left operand is a matrix, we'll make this operator a member of the `Matrix` class; its header has the line:

```
Vector operator*(Vector b);
```


The code from `Matrix10.cpp` is given below.

```
84 // Overload the operator multiplication (*) for a Matrix-Vector
// product. Matrix is passed implicitly as "this", the Vector is
86 // passed explicitly. Will return v=(this)*u
Vector Matrix::operator*(Vector u)
88 {
    Vector v(N); // v = A*u, where A is the implicitly passed Matrix
90     if (N != u.size())
        std::cerr << "Error: Matrix::operator* - dimension mismatch"
92         << std::endl;
    else
94         for (unsigned int i=0; i<N; i++)
            {
96                 double x=0;
98                 for (unsigned int j=0; j<N; j++)
100                     x += entries[i*N+j]*u.geti(j);
102                 v.seti(i,x);
            }
    return(v);
}
```

Check that
this
makes
sense.

10. Extras

We've covered enough about operator for have functioning *Vector* and *Matrix* classes.

However, there is much more that could be done. Have a look at the “Extras” for Week 10, to see some more examples (different classes; different methods).

11. Preview of Lab 8

In Lab 7 you had to write a function that implemented the **Jacobi** method. For Lab 8, you'll rewrite it using the **Vector** and **Matrix** classes.

In addition, you'll implement the Gauss-Seidel method. For that, you'll have to implement the “backslash” operator, for solving $Ax = y$, where A is triangular.

Finished here Friday